UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

The advancement of technology through different computer generations has focused on improving performance, speed, and efficiency. From vacuum tubes in the first generation to modern AI-powered systems, each generation introduced faster processing, smaller components, and greater reliability.

a) Name the inventor of the following devices:

   i. Mechanical Calculating Machine

   ii. Analytical Machine iiD IAS Machine

b) Why are the machines of later computer generation designed for general purposes instead of the eadier computer generation that is more specific in function?

QUESTION A2 **(5 marks)**

Convert 12.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A3 **(5 marks)**

Given the following Figure 2.

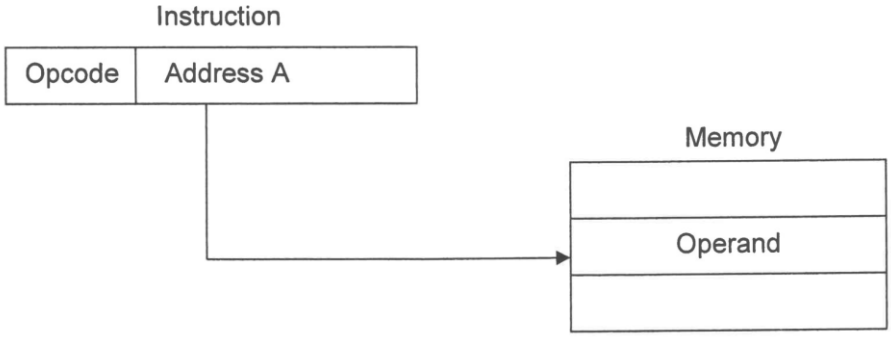


Figure 2

a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

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QUESTION A4 **(5 marks)**

a) List TWO (2) advantages of SMP organization.

b) Define cluster organization.

c) Give ONE (1) benefit of cluster organization.

QUESTION A5 **(10 marks)**

Keyboard is the most common means for computer interaction.

a) Explain TWO (2) types of keyboard characters.

b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

QUESTION A6 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert 291010 to hexadecimal.

b) Convert 29016 to decimal.

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QUESTION A7 **(10 marks)**

a) List FOUR (4) types of access method for storage device with ONE (1) example for each of them.

b) Calculate the total of memory location using 11 address bit.

c) Calculate the address bits required for 2MB of RAM.

QUESTION A8 **(10 marks)**

a) Discuss TWO (2) types of bus timing.

b) Describe THREE (3) Central Processing Unit (CPU) components.

QUESTION A9 **(10 marks)**

Question 7 is based on Figure 2.

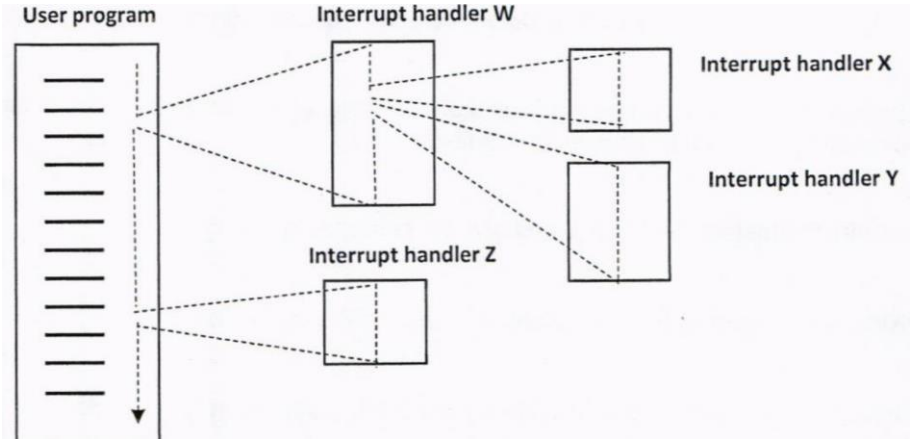


Figure 2: Nested interrupts

Define intenupt.

a) Identify the intemrpt that has the highest priority with relevant justification

b) Identify the interrupt that has the lowest priority with relevant justification

c) Rank the sequence of the intem.rpt process completed ff the computer system applies the

d) defined priority concept to handle multiple intemrpts. Rank the sequence of the interrupt process completed if the co mputer system aPPlies the

e) disable nested interrupt concept to handle multiple interrupts'

QUESTION A10 **(10 marks)**

a) List the phases in the machine cycle,

b) Discuss the function of the different CPU components in the machine cycle.

c) What will happen in the machine cycle if access to the main memory is blocked?

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		